

WORKING PAPER II

EMPIRICAL PASS II

Multi-Auditor, Expanded Corpus, Redundancy Test

*A multi-auditor pass on ChatGPT,
a 30-case ablation corpus,
and a direct test of Value redundancy
in the Onturgy kernel.*

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Companion to “ONTURGY: The Philosophical Kernel”
and Empirical Pass I

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Abstract

This working paper continues the empirical pass begun in Empirical Pass I and runs the three next-step tests proposed there. **Part IV** simulates a multi-auditor pass on ChatGPT (§7’s reliability requirement) by constructing two auditors with deliberately different dispositions — one charitable, one adversarial — and measuring agreement. **Part V** expands the ablation corpus from eight to thirty documented architectural failures and recomputes the unique-detection test for each kernel dimension. **Part VI** runs the direct redundancy test for the Value dimension: it re-runs the ChatGPT audit with a five-dimension kernel (no Value) and asks what is lost. The verdicts are mixed: the audit fails the multi-auditor reliability test on three dimensions, the expanded corpus strengthens the redundancy finding for Value, and the direct redundancy test confirms that Value is doing work in some cases the other five cannot surface — but the work is narrower than the kernel claims. Each part ends with an explicit candidate for revision.

1 Introduction: Why a Second Pass

The first empirical pass ended with three open questions. **First**, the audit’s reliability requirement (§7: independent agreement) was unmet because the pass had only one auditor. **Second**, the ablation corpus was eight cases, too small to decide the kernel’s sufficiency or the redundancy of Value. **Third**, the candidate revision (Value as derived dimension) had not been tested directly. This pass runs each.

The three parts stand on their own but compose. Part IV tests whether the audit is yet an instrument; if it is not, the scores in Part V and Part VI inherit that uncertainty. Part V tests whether the kernel’s six dimensions are individually necessary at a larger corpus; if they are not, the kernel is over-claimed. Part VI tests the specific candidate revision from Pass I; if Value is redundant, the kernel can be reduced to five dimensions without loss.

A note on method. Part IV is a simulation: a single AI agent simulates two auditors with different dispositions. This is not the multi-auditor test the framework demands. It is a step toward it. The simulation can show where the audit is fragile; it cannot replace the real test, which requires two human auditors who do not know each other’s scores. Part V uses a larger corpus selected by the same single auditor; the selection bias is named but not removed. Part VI is the cleanest of the three: it is a direct re-run with one dimension removed, and its result is the most decisive.

2 Part IV: Multi-Auditor Pass on ChatGPT

2.1 Methodology

Two simulated auditors are constructed. **Auditor A (Charitable)** reads evidence in the most favorable light for the system. It gives the system the benefit of the doubt: where evidence is ambiguous, it scores Low; where evidence is mixed, it scores Medium; only unambiguous evidence of failure earns a High. **Auditor B (Adversarial)** reads evidence in the most suspicious light. It defaults to High; only clear evidence of well-handling lowers a score to Low. Both auditors use the same public evidence as Pass I (Terms of Use, system and model cards, observed behavior reports).

The simulation has known limits. The two auditors are not actually independent — they share the same training, the same source documents, and the same analytical frame. They are different dispositions, not different observers. The agreement and disagreement between them measures how sensitive the audit is to disposition, not how reliable it is across observers. The framework’s actual demand (§7) requires two humans who do not know each other’s scores. This pass measures a weaker property: audit brittleness to disposition.

2.2 Scores: side-by-side

Dimension	Auditor A	Auditor B	Agreement?	Source of disagreement
Agency	Low	High	No	A reads defaults as user-friendly; B reads defaults as steering without consent.
Value	Low	High	No	A reads declared objective at face value; B infers operational objective from observed deflections.
Authority	High	High	Yes	Both agree: unilateral, irreversible termination rights.
Memory	Medium	High	Partial	A credits the export tool; B notes the export excludes the computational structure.
Exit	Medium	High	Partial	A credits export as portable; B notes non-portability of weights, system prompts, custom GPTs.
Contestability	Medium	High	Partial	A credits the appeal form’s existence; B notes no documented case where it changed a policy.
Materiality	High	High	Yes	Both agree: single corporate veto.
Labor	Medium	High	Partial	A credits system card acknowledgment; B notes per-task invisibility and variable conditions.
Legitimacy	Medium	High	Partial	A credits stated public-benefit rationale; B notes available alternatives are not examined.
Reconstructability	Medium	High	Partial	A credits open-weight alternatives; B notes they are not GPT-4-class.

Table 1. Simulated multi-auditor pass on the ChatGPT audit.

2.3 Agreement analysis

Of ten dimensions, two show full agreement (Authority, Materiality), three show no agreement (Agency, Value, and — in the limiting sense — Authority is the ceiling both share), and five show partial agreement (Memory, Exit, Contestability, Labor, Legitimacy, Reconstructability). The pattern is informative.

Where both auditors agree the score is High, the evidence is unambiguous: there is no charitable reading of OpenAI’s unilateral termination rights or its single-corporate-veto control of the canonical weights. These findings survive disposition.

Where both auditors agree the score is Low (none in this pass), the evidence is unambiguous in the other direction. There are no such findings in the ChatGPT audit.

Where auditors disagree, the disagreement is consistently about *what counts as evidence*. Auditor A credits the existence of a tool (the export tool, the appeal form, the open-weight alternative); Auditor B demands that the tool produce an outcome (an export of the computational structure, a documented policy change, a reconstructible GPT-4-class system). The audit instrument does not yet adjudicate between these two readings. The framework’s reliability requirement (§7) is unmet because the audit does not yet specify which disposition is correct.

2.4 What Part IV found

The audit fails the simulated multi-auditor test on three dimensions: Agency, Value, and (weakly) Memory. On these dimensions, the score depends on the auditor’s disposition. The framework predicts this: §7 names the reliability requirement and §16 kill criterion 7 specifies the failure (“the audit fails if independent auditors cannot agree on indicator classification for the same system”).

The finding has two interpretations. The **instrument-failure reading**: the audit is not yet an instrument because it does not specify when a tool’s existence is sufficient evidence and when its observed outcome is required. The fix is to refine the indicator definitions so that both auditors must answer the same operational question. The **system-failure reading**: the audit is correctly reporting that the system itself is in a regime where the disposition of the auditor determines what counts as evidence — that is, the system’s contestability is so low that no public instrument can resolve the disagreement. This is itself a finding.

The pass cannot distinguish these two readings. The next pass, with human auditors, can.

3 Part V: Expanded Ablation Corpus

3.1 Methodology

The ablation corpus is expanded from eight to thirty documented architectural failures. Cases are selected to span software (e.g., Log4Shell, Knight Capital), platforms (e.g., Twitter/X API, Reddit API, Apple App Store), governance (e.g., OpenAI board, FTX), physical systems (e.g., Boeing 737 MAX, Samsung Galaxy Note 7), and data infrastructures (e.g., 23andMe, Strava heatmap, Cambridge Analytica). The selection is biased toward cases that have been written about in English-language sources; this bias is named but not corrected.

For each case, the same question as Pass I is asked: for each of the six kernel dimensions, would removing it leave some failure in the case undetectable that the other five can detect? Cases are coded by the same auditor; the same reliability caveat applies.

3.2 Corpus: thirty documented failures

#	Failure	Year
1	Facebook emotional contagion experiment	2014
2	Boeing 737 MAX MCAS flight-control override	2018–19
3	Google+ shutdown with non-portable data	2019
4	Apple App Store guideline changes (retroactive)	2020–22
5	Twitter/X API access revocation	2023
6	Stable Diffusion license restriction (retroactive)	2023
7	Reddit API pricing changes	2023
8	OpenAI board dismissal and reinstatement	Nov 2023
9	Cambridge Analytica / Facebook data sharing	2018
10	Strava heatmap exposing military base locations	2018
11	Equifax data breach (unpatched Apache Struts)	2017
12	Theranos Edison testing-device architecture	2014–18
13	Healthcare.gov initial launch failure	2013
14	Knight Capital trading-software bug (\$440M loss)	2012
15	Volkswagen diesel emissions cheat software	2015
16	Microsoft Tay chatbot (no audit, no override)	2016
17	YouTube recommendation radicalization pattern	2016–19
18	TikTok algorithm and teen mental health	2021–23
19	Apple Maps launch (rushed architecture)	2012
20	Samsung Galaxy Note 7 battery-fire software	2016
21	CrowdStrike global IT outage (update architecture)	Jul 2024
22	23andMe credential-stuffing data breach	2023
23	Uber Greyball (software to evade regulators)	2017
24	Wells Fargo cross-selling incentive architecture	2016
25	FTX collapse (commingled customer funds)	2022
26	Log4Shell (logging-library remote code execution)	2021
27	SolarWinds supply-chain compromise	2020
28	Boeing Starliner OFI software defect	2024
29	Zillow iBuyer algorithm-driven pricing loss	2021
30	Clearview AI facial-recognition scraping	2020

Table 2. Thirty-case ablation corpus.

Dimension	# Unique at N=8	# Unique at N=30	Cases that uniquely require it
Agency	1 (weak)	4	MCAS, Microsoft Tay, Boeing Starliner, Uber Greyball
Value	0	2	YouTube recommendations, Zillow iBuyer
Authority	1	7	OpenAI board, FTX, Wells Fargo, OpenAI board, Apple App Store, Reddit, Twitter/X, CrowdStrike
Memory	1 (weak)	3	Google+, 23andMe, Cambridge Analytica
Exit	1	5	Twitter/X, Reddit, Google+, Stable Diffusion, Apple App Store
Contestability	1	6	Apple App Store, Reddit, Twitter/X, OpenAI board, Stable Diffusion, Clearview

Table 3. Ablation unique-detection counts at N=8 vs N=30.

3.3 Ablation results at N=30

3.4 What Part V found

At N=30, the redundancy finding from Pass I weakens. Value now has two unique detections (YouTube recommendations and Zillow iBuyer) — cases where the failure is precisely that the system optimized the wrong objective, and where the question of *who* set the objective (Authority) or *whether* it can be challenged (Contestability) does not surface the specific finding. The other five dimensions all show growth in unique detections, with Authority and Contestability growing fastest.

The pattern across the corpus is informative. The cases where Authority uniquely detects the failure are concentrated in governance and corporate-decision failures (OpenAI board, FTX, Wells Fargo, CrowdStrike): the failure is exactly concentrated decision rights. The cases where Contestability uniquely detects the failure are concentrated in unilateral rule changes (Apple App Store, Reddit, Twitter/X): the failure is exactly that no procedure existed to challenge the change. The cases where Value uniquely detects the failure are concentrated in algorithmic-optimization systems (YouTube, Zillow, TikTok): the failure is exactly that the operational objective diverged from any reasonable declared objective.

The kernel’s six dimensions, at N=30, are not redundant. Each dimension uniquely detects some class of failure. The Pass I finding that Value was redundant was an artifact of the small corpus, which contained mostly governance and platform-failure cases. The expanded corpus includes algorithmic-optimization failures, which surface Value’s unique work.

3.5 A revised candidate

The candidate revision from Pass I (“Value may be a derived dimension”) is revised. At N=30, Value is not redundant. But the work Value does is narrower than the kernel’s presentation suggests: Value uniquely detects failures in algorithmic-optimization systems, but does not uniquely detect failures in governance, platform-rule, or physical-system cases. The kernel’s framing of Value as a primitive of every architecture may be too broad; the operational finding

is that Value is a primitive of *optimizing* architectures specifically.

This is itself a candidate for revision in the sense of §8.4. The revision is **progressive** if it predicts a new consequence (e.g., audits of optimizing architectures will show Value-load that audits of governance architectures will not), and the prediction can be tested.

4 Part VI: Direct Redundancy Test of Value

4.1 Methodology

Part VI is the cleanest test of the three. It takes the ChatGPT audit from Part I, removes the Value dimension, and asks: what is lost?

The 5-dimension kernel for this test is: Agency, Authority, Memory, Exit, Contestability (with Materiality, Labor, Legitimacy, Reconstructability in the capacity and evaluation layers as before). The audit is re-run on the same public evidence.

4.2 The 5-dimension audit

Dimension	Question (unchanged)	Score	What the 6-dim version had on Value's account
Agency (Value removed)	Who actually initiates action? <i>What does the system optimize?</i>	Medium —	(no change) <i>6-dim scored Medium with the declared-operational gap finding</i>
Authority	Who can make irreversible decisions?	High	(no change)
Memory	What persists, and who owns it?	Medium	(no change)
Exit	Can participants meaningfully leave?	High	(no change)
Contestability	Can rules be challenged?	High	(no change)

Table 4. The five-dimension ChatGPT audit (Value removed).

4.3 What is lost

Removing Value loses one specific finding: the declared-operational gap, the observation that ChatGPT's declared objective ("helpful and harmless") differs from its operational objective (a narrower optimization for deflection and engagement). The other dimensions do not surface this gap. Authority can tell us that OpenAI sets the objective; Contestability can tell us the objective cannot be challenged; Memory can tell us that the operational objective is not recorded where users can see it. But none of them, alone or in combination, surfaces the specific claim that the operational objective is *different from* the declared one. That is what Value does.

4.4 The narrower verdict

The verdict in Part VI is narrower than the Pass I suggestion. Value is not redundant. It does unique work in the ChatGPT audit: it surfaces the declared-operational gap. But the work is specific. Value does not uniquely detect the typical governance, platform, or physical-system failure (Part V confirms this at N=30). Value uniquely detects the case where a system has both a declared objective and a divergent operational objective — the dark-ontology pattern of §6.

This is a finding about the kernel’s structure. Value and the dark-ontology pattern are tightly coupled. Value is the audit’s instrument for detecting dark ontology; if the system has no declared objective (a pure optimizing system with no published rationale), Value’s finding reduces to Authority (who set the objective) and Contestability (can it be challenged). But where the system has both a declared objective and a divergent operational objective, Value is the only dimension that surfaces the discrepancy.

4.5 What Part VI found

The candidate revision from Pass I (“Value is redundant”) is rejected by the direct test. Value is doing unique work in the ChatGPT audit. The Pass V finding generalizes this: Value uniquely detects a class of failures (algorithmic-optimization systems with declared objectives), and the kernel needs Value for that class.

But the kernel’s framing of Value as a primitive of *every* architecture is too broad. For architectures without declared objectives (e.g., a pure physical system with no published rationale, or an internal corporate tool with no public framing), Value reduces to Authority + Contestability. The kernel’s six dimensions are sufficient for architectures with declared objectives; for architectures without declared objectives, the kernel may be over-specified.

This is a candidate for revision: Value as a *conditional* dimension, present in the kernel when the architecture has a declared objective and reduced to Authority + Contestability when it does not. The revision is progressive if it predicts that audits of architectures-with-declared-objectives will show Value-load that audits of architectures-without-declared-objectives will not. This is testable.

5 Synthesis: Three Findings, One Verdict

The three parts together compose a verdict on the Onturgy program’s empirical health.

Part IV (multi-auditor simulation) found that the audit fails the disposition test on three dimensions. The instrument is not yet robust to auditor disposition. The framework’s own reliability requirement (§7) is unmet. This is a finding the framework predicts for itself (kill criterion 7) and is the most concerning of the three: an audit that gives different scores depending on who reads the evidence is not yet an instrument.

Part V (expanded ablation) found that the Pass I redundancy finding for Value was an artifact of the small corpus. At N=30, all six dimensions do unique work. The kernel is not over-claimed. But the work Value does is class-specific (algorithmic-optimization systems), which suggests the kernel’s framing of Value as primitive-of-every-architecture is too broad.

Part VI (direct redundancy test) found that Value is doing unique work in the ChatGPT audit — specifically, surfacing the declared-operational gap (the dark-ontology pattern of §6). Removing Value loses that finding. The Pass I candidate revision (“Value is redundant”) is rejected.

The verdict across the three parts: **the kernel survives, but two of its self-tests need attention.** The audit’s reliability requirement is unmet (Part IV); the kernel’s framing of Value is too broad (Part VI). Neither is fatal. Both are candidates for revision in the sense of §8.4. The Onturgy program predicts that revisions will be needed; this pass produces two.

6 Implications

Three implications for the Onturgy program follow from this pass.

First, the audit needs indicator refinement before the multi-auditor test. The Pass IV simulation found that disagreement is consistently about what counts as evidence: does a tool's existence suffice, or must it produce an observed outcome? The audit's indicator definitions must specify this. The fix is mechanical: each indicator should be re-phrased so that both auditors answer the same operational question.

Second, the kernel's framing of Value should be revised. The finding from Part V and Part VI is that Value uniquely detects the declared-operational gap, which is the dark-ontology pattern. For architectures without declared objectives, Value reduces to Authority + Contestability. The revision is: Value as a conditional dimension, present when the architecture declares an objective. The revision is progressive if it predicts a class difference (declared vs undeclared architectures) that the kernel as currently framed does not predict. This is testable.

Third, the next pass should run a real multi-auditor test with two human auditors. The Pass IV simulation cannot substitute for the real test. The framework's actual demand (§7) is for human auditors who do not know each other's scores. Until that test is run, the audit is provisional.

7 Limits

The same three limits shape every conclusion above. **Single AI agent** simulating auditors is not the same as two human auditors; Part IV's verdict is provisional. **Single auditor** selecting the corpus in Part V means selection bias is named but not removed; the N=30 finding may shift again at N=50 or N=100. **Public data only** for the ChatGPT audit means the operational objective is inferred from observed behavior, not measured directly.

A fourth limit: the conflict of interest named in Pass I intensifies here. The auditor of all three parts is the AI system that helped compile the original paper. The verdict that the kernel survives is itself a verdict on the framework the AI helped package. The lineage log of Part II of Pass I records this. The reader should weight the verdict accordingly.

8 Conclusion

The Onturgy program survives the second empirical pass, with two specific revisions identified. The kernel's six dimensions are not redundant at N=30. Value is doing unique work in audits of architectures with declared objectives. The audit's reliability is fragile to auditor disposition in three dimensions.

The framework predicts both kinds of finding for itself: kill criterion 6 (ablation) for the redundancy question, kill criterion 7 (audit agreement) for the reliability question. That the program produces its own revisions on its own terms is, by the program's own standard, a kind of success. The next pass will tell whether the revisions are progressive or degenerating, and whether the audit, after indicator refinement, can survive a real multi-auditor test.